TEMPORAL CODE CONSISTENCY AND NORMAL-CODE REPAIR FOR RECONSTRUCTION-BASED WASTE ANOMALY LOCALIZATION

ABSTRACT

Reconstruction-based waste detection in an unsupervised setting requires localizing illegal objects in a heterogeneous visual stream based on only normal image training. We expand a trained hierarchical vector-quantized autoencoder architecture by means of three additional objectives. First, we apply a temporal consistency criterion between bottom codes in the current frame and detached bottom codes in the previous normal frame. Then, we use a frequency gated criterion where rarely appearing bottom codes are replaced by most common normal code, resulting in the blend of RGB normal and repaired reconstructions. Finally, we perform a decoder-only fine-tuning on learning how to reconstruct images along the two latent paths. In terms of 496 images labeled as waste, the chosen archiving technique increases mean IoU from 0.33782 to 0.36135 without the increase of network parameters. Our comparisons consist of compact RGB autoencoders and other contemporary reconstruction approaches, using native results only.

Keywords - Anomaly localization, waste sorting, vector quantization, temporal consistency, reconstruction

1. INTRODUCTION

Waste sorting classification requires the detection of a semantic anomaly localization task, since the waste material is undesirable for the sorting line even though it appears visually normal. WS2 collects a diverse video stream of the sorted waste both before and after the removal of the undesirable objects [1]. After frames mainly show desirable waste materials despite potential contamination. However, they do not require a classifier on the undesirable objects or a labeled dataset of individual pixels to learn from. This project adopts after frames to train the model and before frames to validate.

The fundamental problem is that faithful reconstruction using autoencoders and anomaly detection using selective anomaly failure represent separate problems. With proper training, an autoencoder can faithfully reconstruct all sorts of objects using familiar colors, edges, and textures; merely by minimizing normal reconstruction error, one cannot guarantee that any anomalous material is granted a large residual. Rossi's reconstruction paradigm [2] uses a hierarchy of VQ-VAE-2 together with full-image and crop-based objectives. We adapt its learned reconstruction model while discarding SAM scoring and refinement.

The research question is whether temporal consistency in normal frame and a conservative approach toward reconstructing rare discrete symbols will enhance foreground localization without the requirement for additional neural resources. The idea here is that the use of constraints on the bottom representation for the set of normal consecutive frames and the constrained way to rebuild the rare codes leads to a higher mean frame IoU compared to inherited autoencoder.

The final compact framework tests three things. Firstly, effects of temporal consistency introduced during training. Secondly, whether replacing rare codes with the most common code help in finding anomalies and finally, how decoder only training can help in improving the response beyond frozen repair. Using this compact methodology, our model reaches mean frame of IoU 0.36135 versus 0.33782 compared to Rossi's model [2]. It should also be noted that this gain came with no added neural parameters.

2. RELATED WORK

The kind of information that the autoencoder saves depends on the error left available to be detected. The structural similarity training [3] process alters this principle from the usual pixel accuracy, while discrete bottleneck representation [4, 5] reduces the representation through dictionaries. None of these two principles alone guarantees the joint distribution density of normal code grids. Memory based sparse addressing [6] imposes an extra constraint in terms of learning normal prototypes. The problem is that useful details of the normal state and anomalies might be represented similarly. In our study, we apply the second frequency principle to the already established codebook. We propose a frequency-based rule on an existing codebook to address this issue.

Restoration techniques and inpainting [7, 8] limit the data available to the neural network prior to reconstruction, and anomaly injection and latent perturbation techniques [9, 10] affect how corruptions are handled. On the other hand, feature scaling [11] acts on the process of decoding. Both approaches tackle the issue of copying the identity at different stages of the process through different types of corruption, training objectives, and heads for detection. Our question is more specific as we investigate whether intervention on bottom level codes can affect an already trained decoder's output while an explicit blend limits deviation from its ordinary reconstruction? Decoder-only adaptation tests whether learning the intervention's response adds value beyond applying it to frozen weights.

The feature-reconstruction detector [12, 13] infers the descriptions with the help of a pretrained frozen encoder, while an RGB autoencoder predicts intensity information. For a Q-Former image reconstruction approach [14], the pretrained perceptual scorer can additionally be employed. Both types of models have a localization interface, and residuals in each case reflect different kinds of differences and are transferable through external representation learning [15]. Therefore, the test set common for both models will be required, but insufficient for understanding the rankings – the native scores and scoring pretraining will still need to be observed. We compare RGB outputs under shared residual scoring and retain feature maps.

The video-memory techniques [16, 17] pass on data from frame to frame; however, this does not mean that normal correspondences cannot also play a part in restricting the representation learning process. **Inspired also by a supplied unpublished cross-attention manuscript [18], we use detached previous-frame codes only as training targets, retaining single-frame inference.** The frequency provides a supplementary limited concept of normality, where rare codes can be found in the desired content, and familiar codes can represent illegal content. In addition, the distribution shift research [19] results caution us from conflating probability with anomaly semantics. The proposed solution operates on frequencies empirically, but does not pretend to approximate the entire normal distribution of images.

3. METHOD

We initialize the inherited VQ-VAE-2 from its trained checkpoint (Fig. 1). Temporal adaptation updates the existing backbone using the inherited objective plus temporal loss (Fig. 2). Normal-code repair supplies altered decoder inputs (Fig. 3). A separate stage freezes upstream modules and trains only the inherited final decoder (Fig. 4). The selected model uses bounded repair at inference. The matching module adds no neural parameters.

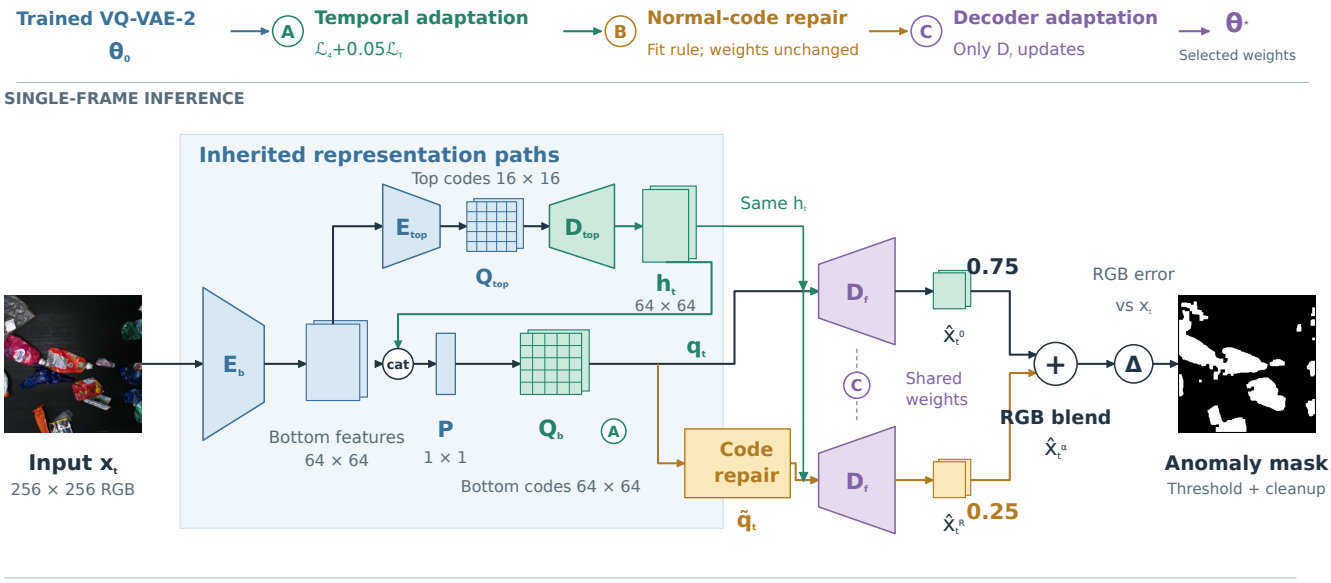


Fig. 1: Architectural overview of the proposed model

3.1. Hierarchical reconstruction backbone

For a 256×256 RGB frame x_t, the bottom encoder produces a $64 \times 64 \times 64$ tensor. A second encoder and top codebook produce a $16 \times 16 \times 64$ top representation. Decoded top context h_t is concatenated with bottom encoder features and projected to the bottom codebook input. Both codebooks contain $K = 1024$ vectors of dimension 64 and use EMA updates. The final decoder receives the quantized bottom representation q_t and top context:

$$q_t = Q_b(P[E_b(x_t), h_t]) \tag{1}$$

$$\hat{x}_t^0 = D_f(q_t, h_t) \tag{2}$$

The sigmoid image output lies in $[0,1]$. The backbone has 2,604,483 registered parameter entries, including EMA codebooks. All proposed components reuse these entries.

The original adaptation objective combines a resized full-frame view, four corner crops of a 512×512 training frame, crop/full consistency, and VQ commitment terms:

$$\mathscr{L}_4 = \mathscr{L}_{\text{full}} + \mathscr{L}_{\text{crop}} + \mathscr{L}_{\text{cons}} + \mathscr{L}_{\text{VQ}} \tag{3}$$

3.2. Training-only bottom-code temporal consistency

Figure 2 expands the training-only temporal branch. For each current location p, previous-frame bottom codes at valid offsets $d \in [-4,4]^2$ form the candidate set. Candidate similarity is

$$s_{p,d} = \cos(\text{sg}(q_t(p)), \text{sg}(q_{t-1}(p+d))) - 0.04\,\frac{\|d\|_2^2}{16} \tag{4}$$

Let $a_{p,k}$ be the softmax of the two retained scores, with temperature 0.08. The propagated target is

$$m_p = \sum_{k=1}^{2} a_{p,k}\, q_{t-1}(p + d_{p,k}) \tag{5}$$

The implemented legacy confidence is entropy-based:

$$H(a_p) = -\sum_{k=1}^{2} a_{p,k} \log a_{p,k} \tag{6}$$

$$c_p = \left(1 - \frac{H(a_p)}{\log 2}\right) \text{sigmoid}\left(\frac{s_{p,1} - 0.35}{0.10}\right) \tag{7}$$

Thus, the additional temporal consistency loss is

$$\mathscr{L}_T = \frac{\sum_p \text{sg}(c_p)\left[1 - \cos\left(q_t^{\text{ST}}(p), \text{sg}(m_p)\right)\right]}{\sum_p \text{sg}(c_p) + 10^{-6}}. \tag{8}$$

The straight-through current code preserves a gradient to the current encoding/pre-quantization path. Previous codes are computed without gradients or EMA updates. Candidate matching and target construction are detached. The final image decoder receives no direct gradient from $\mathscr{L}_T$, while the ordinary reconstruction terms still train it. Adaptation minimizes $\mathscr{L}_4 + 0.05\mathscr{L}_T$. The temporal term is applied only to the resized full-frame pass, not the four crop passes. Only the previous frame is used ($T = 1$), and first frames are excluded from both temporal and matched continuation training.

3.3. Normal-code repair and bounded reconstruction

The temporal objective constrains training but leaves ordinary inference unchanged. Figure 3 shows the subsequent intervention on normal-code rarity. After adaptation, a histogram counts bottom code IDs over normal training frames. With Laplace smoothing,

$$\hat{p}(k) = \frac{n_k + 1}{\sum_{j=1}^{K} n_j + K}, \qquad r(k) = -\log \hat{p}(k) \tag{9}$$

Let τ_r be the higher empirical 99th percentile of code rarity occurrences in normal selection frames. The strict gate is $g_p = \mathbf{1}[r(k_p) > \tau_r]$, evaluated in float64 to preserve ties. At gated locations, the code is replaced by $e_{k^\star}$, where $k^\star = \arg\max_k n_k$; other bottom codes and all top context remain unchanged. This is a global unigram rule, not a conditional spatial prior or joint image-density model.

Using the same final decoder, the repaired latent produces $\hat{x}_t^R = D_f(\tilde{q}_t, h_t)$. The scored reconstruction is

$$\hat{x}_t^\alpha = (1-\alpha)\hat{x}_t^0 + \alpha \hat{x}_t^R, \quad \alpha = 0.25. \tag{10}$$

The blend bounds output intervention. Inference needs one shared encoding and two calls to the existing final decoder, with no previous-frame input.

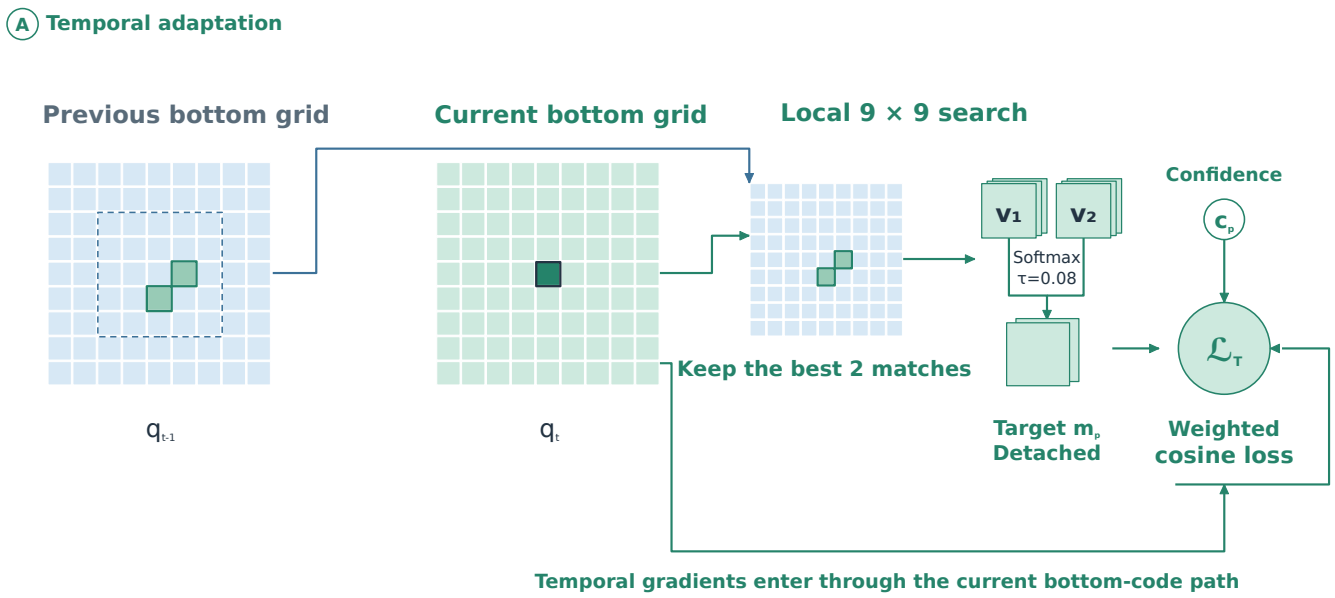


Fig. 2: Training-only temporal bottom-code consistency architecture.

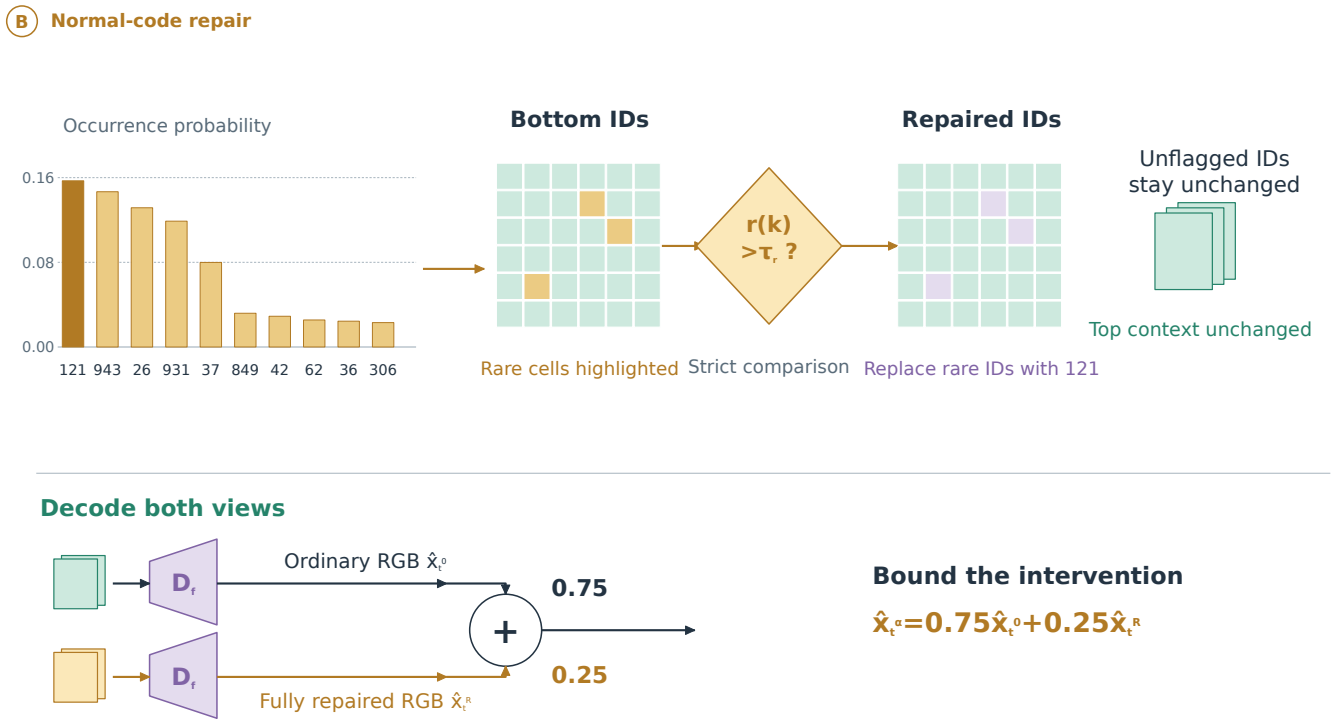


Fig. 3: Normal-code repair and bounded reconstruction framework

3.4. Repair-aware final-decoder adaptation

Repair can be applied directly to a frozen QMem checkpoint. The selected 0.36135 variant instead adds the decoder-only adaptation shown in Fig. 4. This separate second training stage is initialized from QMem. Encoders, codebooks, pre-quantization projection and top decoder remain frozen. Only D_f is optimized using normal images:

$$\mathscr{L}_D = \frac{\mathrm{MSE}(x,\hat{x}^0) + 0.5\,\mathrm{MSE}(x,\hat{x}^R)}{1.5}. \tag{11}$$

This stage trains with the full repaired path, while inference uses the bounded blend in Eq. 10. Normal selection favors low bounded-reconstruction MAE subject to ordinary MAE remaining within 5% of the initial QMem value. This is not a simultaneous four-loss, temporal and repair objective. A matching ordinary-decoder adaptation control is retained in the supplement. No neural parameters are added by this stage.

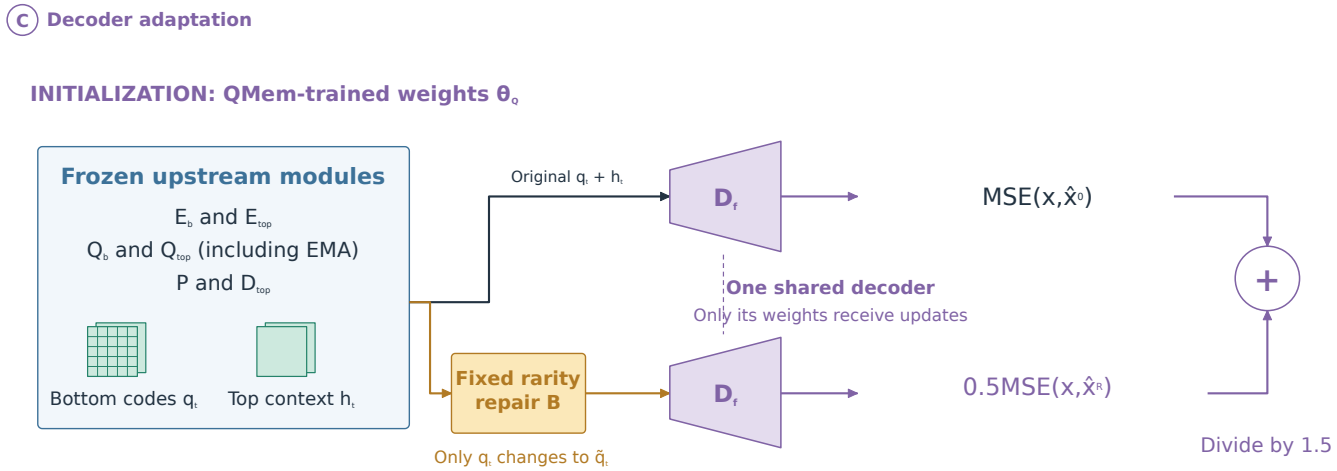


Fig. 4: Repair-aware final-decoder adaptation framework

Table 1: Archived staged ablation. H: historical mean frame IoU; N: normal-calibrated target-2% IoU. Pixel AP uses the continuous RGB error map.

Variant	H	N	AP
Inherited reference	0.3378	0.2358	0.5942
QMem	0.3551	0.3419	0.4601
QMem + bounded repair	0.3592	0.3503	0.4844
Decoder-aware + repair	0.3613	0.3116	0.4529

4. EXPERIMENTAL PROTOCOL

4.1. Data, scoring and metrics

Evaluation separates normal-data fitting and selection from annotated before-frame testing. We use the existing WS2-ECA split: 3,829 after frames for training, 883 after frames for validation, and 496 annotated before frames from 14 videos for testing. The validation pool is partitioned into 521 selection, 128 calibration and 234 normal-audit frames. No test masks enter gradient training or within-run checkpoint selection. After frames are treated as intended normal observations rather than guaranteed perfectly clean material.

For RGB models, the shared score is the channel-mean absolute reconstruction error smoothed with a Gaussian of $\sigma = 1$. Historical Thresholding uses all 883 normal validation frames and $\tau_H = Q_{0.75} + 4(Q_{0.75} - Q_{0.25})$. A second protocol uses a prespecified 2% raw-pixel false-positive target estimated from 4,096 pixels per calibration frame. Achieved postprocessed FPR is measured on the normal-audit partition and is reported, not assumed equal to 2%.

All prediction masks undergo the same closing, opening, hole filling and small-component removal. Test masks use identical bilinear resizing to 256×256 followed by a 0.5 threshold. The primary statistic is foreground mean frame IoU,

$$\mathrm{mIoU} = \frac{1}{496} \sum_{i=1}^{496} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}, \tag{12}$$

with IoU defined as 1 for an empty union. This differs from pooled IoU and from a two-class background/foreground mean. We also retain mean frame Dice, pooled precision/recall, pixel AP, AUROC and normalized AUPRO up to FPR 0.30. Image-level AUROC is undefined when all evaluated images contain anomalies; image AP is uninformative in this single-class subset and is not reported. Paired intervals resample complete videos, reflecting frame dependence.

4.2. Comparators and scope

Compact RGB baselines are AE-MSE, AE-SSIM, plain VQ-VAE-2, extended VQ-VAE-2 with four loss, a two-dimensional RGB MemAE adaptation, and FAIR-32 without DTD textures. RIAD is a larger restoration comparator using a third-party reimplementation. Dinomaly and original INP-Former reconstruct frozen pretrained features; QFAE is an externally pretrained image-reconstruction/perceptual-score comparator. Their native error maps are evaluated alongside shared RGB residuals where available. Shared evaluation controls the training, validation and test set along with scoring and postprocessing. The proposed archived variants inherit a trained checkpoint whose upstream budget is unknown; several comparators start from initialization, whereas foundation methods inherit external pretraining. The selected archival experiment also uses one seed and was developed on previously inspected test videos.

5. RESULTS AND DISCUSSION

We first assess the staged extension under the historical protocol, then compare baselines under shared and native scores.

5.1. Selected archived reconstruction results

Table 1 separates temporal adaptation, frozen repair and decoder adaptation. QMem increases historical mean frame IoU from 0.33782 to 0.35509; bounded repair raises it to 0.35924, and decoder adaptation yields 0.36135. The total gain over the inherited reference is 2.352 percentage points. It is accompanied by a change in operating behavior: mean frame recall rises from 0.3872 to 0.7323, while mean frame precision falls from 0.7418 to 0.4213. More annotated foreground is recovered, but more background is flagged. The reference's higher pixel AP shows that this thresholded-IoU improvement is not a general improvement in continuous pixel ranking.

Frozen bounded repair improves both mean frame precision and recall over QMem at their respective historical thresholds, while also raising pixel AP. The decoder stage then trades some recall for precision and adds only 0.210 historical-IoU percentage points; its paired video-bootstrap interval includes zero. Frozen repair retains higher normal-calibrated IoU and pixel AP.

Figure 5 connects these stages to the final historical prediction. Both RGB paths use the repair-adapted decoder; their bounded blend supplies the residual map and thresholded mask. Evaluation-only inference reproduced its archived confusion counts and those of three fixed frames without updating weights or refitting thresholds. Several annotated objects are recovered, while missed fragments and boundary errors remain.

5.2. Baseline comparisons and score dependence

Table 2 combines all retained baselines and archived proposed variants, identifying shared RGB-MAE and native score maps separately. The selected model exceeds AE-SSIM by 0.144 historical-IoU percentage points. Figure 6 shows all model–score entries under historical thresholding on one zero-origin axis, ordered from lower to higher IoU.

A common RGB residual controls the comparison but can differ from a method's intended score. MemAE achieves historical IoU 0.32286 with its native squared-error map, compared with 0.08993 under shared MAE. Reporting shared MAE alone would therefore understate its performance. Table 2 and the supplement retain native scores and normal-calibrated results. Threshold policy is particularly consequential for feature methods: normal calibration gives IoU 0.23718 for Dinomaly and 0.32694 for INP-Former. Their nearly empty historical masks indicate an interaction with thresholding, rather than evidence that feature reconstruction cannot localize unwanted material.

5.3. Interpretation and limitations

The results support an operating-point intervention rather than a complete normality model. Temporal targets constrain agreement of normal bottom codes; repair changes the decoder's latent input; decoder adaptation changes its response. A global histogram cannot distinguish rare wanted material from unwanted material assigned to familiar codes, and unchanged top context can preserve anomalous information.

Beyond this dataset, the reusable design is the separation of normal-only temporal training from single-frame inference and the use of existing codes and decoder weights. In contamination screening, higher recall could be useful where misses are costly, while lower precision requires assessing the cost of false alarms. It needs no anomaly annotations or additional neural parameters for fitting. Repair nevertheless requires two final-decoder passes, and a new domain would require its own normal-code counts and gate calibration. Transfer of the gain has not been tested.

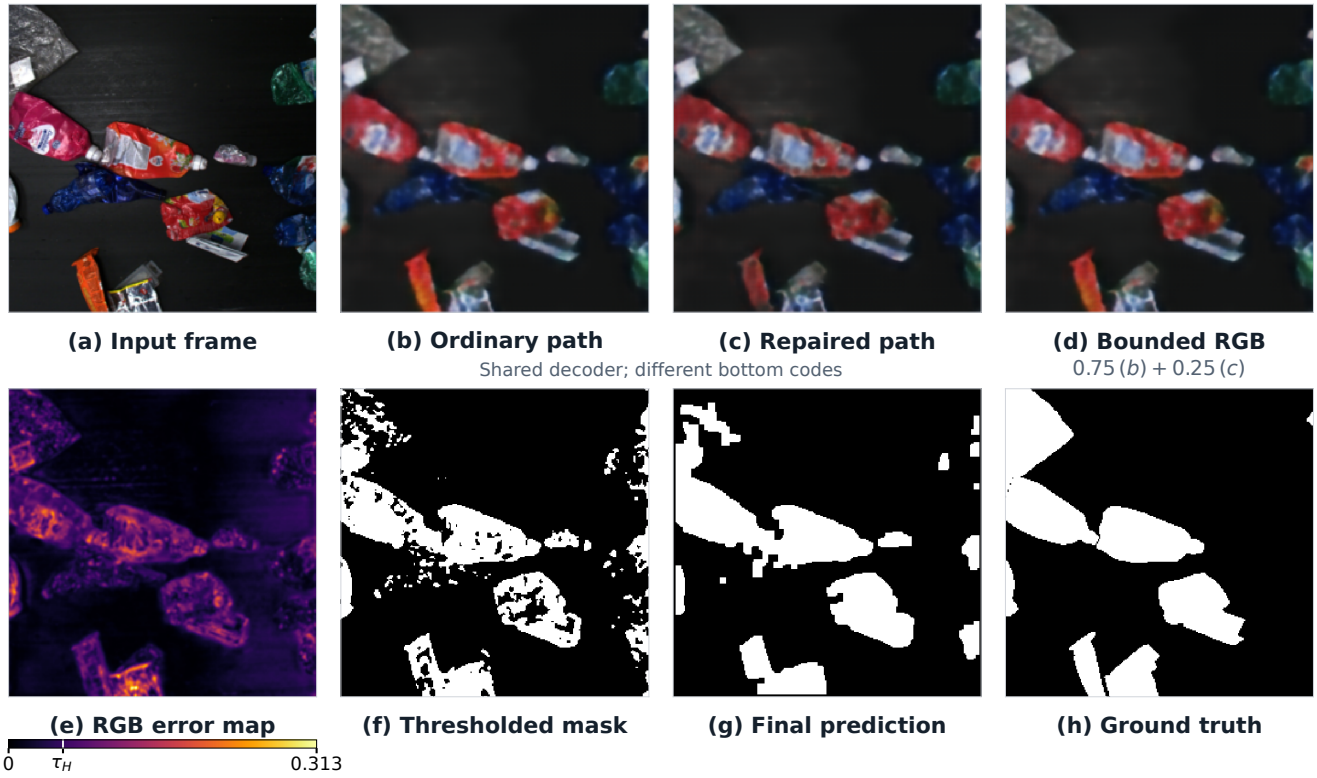


Fig. 5: Historical inference of the selected decoder-adapted model.

Table 2: Unified comparison on the same 496 test frames. Score-map labels distinguish shared RGB MAE from native alternatives. H and N are foreground mean frame IoU under historical thresholding and target-2% normal calibration, respectively. F_N is achieved postprocessed normal-audit FPR (%); AP and AUROC are pixel measures. Parameter counts refer to reconstruction models; QFAE additionally uses a frozen native perceptual scorer (659.42M unique system parameters). Initialization and training exposure differ.

Model	Score map	Params (M)	H IoU	N IoU	F_N	AP	AUROC
AE-MSE	RGB MAE	2.689	0.33325	0.19132	1.82	0.37287	0.79632
AE-MSE	MSE	2.689	0.25572	0.17165	1.67	0.38068	0.81317
AE-SSIM	RGB MAE	2.689	0.35990	0.13396	2.28	0.41462	0.84067
AE-SSIM	SSIM	2.689	0.00943	0.07948	1.40	0.31567	0.76743
Plain VQ-VAE-2	RGB MAE	2.604	0.32767	0.18873	2.36	0.38090	0.83239
Plain VQ-VAE-2	MSE	2.604	0.28669	0.17105	2.34	0.37929	0.83663
MemAE-2D	RGB MAE	2.511	0.08993	0.13493	1.32	0.33144	0.80667
MemAE-2D	MSE	2.511	0.32286	0.11825	1.23	0.33316	0.81321
RIAD	RGB MAE	28.806	0.33076	0.14575	1.56	0.41011	0.84726
RIAD	MSGMS	28.806	0.00016	0.06535	2.32	0.25400	0.75835
FAIR-32, no DTD	RGB MAE	4.322	0.17038	0.14512	2.98	0.26272	0.59252
FAIR-32, no DTD	Native restoration	4.322	0.15581	0.07040	2.97	0.26384	0.74067
Dinomaly	Cosine	148.009	0.00070	0.23718	2.64	0.29724	0.74429
INP-Former	Cosine	155.102	0.00019	0.32694	2.74	0.48119	0.85208
QFAE	RGB MAE	356.122	0.27671	0.23620	2.32	0.41844	0.75556
QFAE	Perceptual	356.122	0.08880	0.21254	4.94	0.30785	0.74117
Christian reference	RGB MAE	2.604	0.33782	0.23579	2.79	0.59423	0.84837
QMem	RGB MAE	2.604	0.35509	0.34194	2.10	0.46006	0.84314
QMem + bounded repair	RGB MAE	2.604	0.35924	0.35032	2.30	0.48437	0.84942
Decoder-aware + bounded repair	RGB MAE	2.604	0.36135	0.31161	2.11	0.45285	0.84224

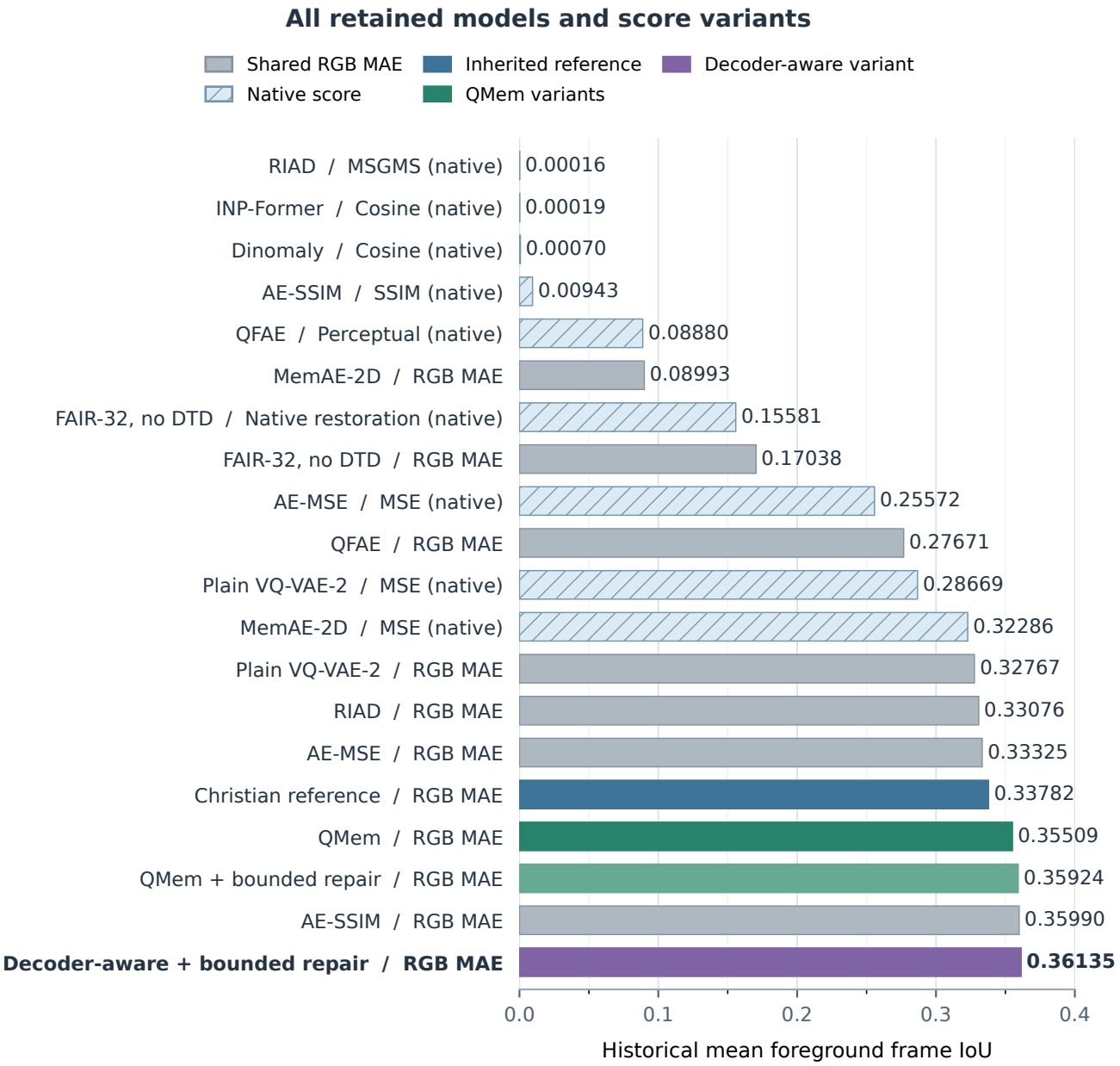


Fig. 6: Historical mean foreground frame IoU for all retained models and score maps on 496 test frames, sorted from lower values at the top to higher values at the bottom. RGB-MAE and native-score entries are labelled separately; all bars use the historical thresholding protocol. Target-2% calibrated results are reported separately in Table 2.

6. CONCLUSION

We extend a trained hierarchical RGB autoencoder through temporal bottom-code consistency, normal-frequency repair and bounded reconstruction blending, with optional final-decoder adaptation. The selected archived model reaches historical mean frame IoU 0.36135 without adding neural parameters. Frozen repair improves both historical and normal-calibrated IoU over QMem; the smaller decoder increment and gap over AE-SSIM remain uncertain. The contribution is a separable mechanism for modifying reconstruction behavior, whose cross-domain gains require independent confirmation.

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